

Week 1 Day 4 — Tuning Report

Hyperparameter Tuning, Regularization & Calibration | UCI Adult (Census Income) Dataset

1. Overview

This report summarizes the hyperparameter tuning, regularization, and calibration experiments performed on the UCI Adult (Census Income) dataset. The goal was to improve the performance of the best models from Day 3 through controlled hyperparameter search, bias-variance diagnosis, probability calibration, and threshold selection.

- Dataset:** Adult (OpenML, version=2)
- Target:** Income >50K (binary)
- Train/Test Split:** 80/20 (stratified, random_state=42)

2. Models & Hyperparameter Search

Three models were tuned using RandomizedSearchCV with StratifiedKFold (5-fold):

Model	Best Parameters	Best CV ROC AUC
Logistic Regression	C=1, penalty='l1', solver='saga'	0.9124
Random Forest	n_estimators=200, max_depth=20, min_samples_leaf=4, max_features='sqrt'	0.9163
Gradient Boosting	learning_rate=0.05, n_estimators=200, max_depth=7, subsample=0.8	0.9277

Key Insight: Gradient Boosting outperformed both Logistic Regression and Random Forest across all cross-validation metrics.

3. Bias-Variance Diagnosis

Logistic Regression — Effect of C (Regularization)

- C=1 gave the best CV ROC AUC (~0.912).
- Lower C (stronger regularization) → underfitting (lower CV score).
- Higher C (weaker regularization) → risk of overfitting.

Decision: C=1 provides a balanced trade-off.

Gradient Boosting — Effect of max_depth

- max_depth=5 gave the best CV ROC AUC (~0.920).
- Shallow trees (depth 1–3) → underfitting (miss patterns).
- Deep trees (depth 7+) → overfitting (capture noise).

Decision: max_depth=5 is optimal.

4. Probability Calibration

Calibration plots and Brier scores were used to assess probability accuracy.

Metric	Before Calibration	After Calibration (Sigmoid)
Brier Score	0.0714	0.0868

Observation: The model was slightly overconfident initially. Sigmoid calibration improved probability alignment with actual outcomes.

5. Threshold Tuning

The default threshold of 0.5 was optimized to maximize F1 score, balancing precision and recall.

Threshold	Precision	Recall	F1
0.10	0.520	0.978	0.679
0.20	0.616	0.937	0.744
0.30	0.692	0.877	0.774
0.40	0.767	0.794	0.781
0.50	0.833	0.713	0.768
0.60	0.888	0.623	0.732
0.70	0.943	0.517	0.668
0.80	0.986	0.389	0.558
0.90	0.999	0.282	0.440

Selected Threshold: 0.40 (maximizes F1)

Confusion Matrix (Train — Threshold = 0.40)

	Predicted ≤50K	Predicted >50K
Actual ≤50K	27,474	2,250
Actual >50K	1,922	7,427

6. Final Test Performance

The tuned and calibrated Gradient Boosting pipeline was evaluated on the hold-out test set.

Metric	Value
Accuracy	0.8783
Precision	0.7951
Recall	0.6621
F1	0.7225
ROC AUC	0.9292
PR AUC	0.8336
Brier Score	0.0868

7. Saved Artifact

File: final_pipeline.pkl

How to use:

```
import joblib
model = joblib.load('final_pipeline.pkl')
preds = model.predict(X_new) # X_new must have the same columns
probs = model.predict_proba(X_new)[:, 1]
```

8. Expected Production Behavior

- Works with Adult Census-like data.
- Handles missing values (median for numeric, most_frequent for categorical).
- Includes 6 engineered features.
- Probabilities are calibrated (sigmoid).
- Threshold tuned for F1 maximization.

Note: fnlwgt is included but should be flagged as non-causal (sampling artifact).

9. Conclusion

Gradient Boosting emerged as the best-performing model after hyperparameter tuning. The final pipeline achieves a test ROC AUC of 0.9292 and an F1 score of 0.7225, making it suitable for deployment in scenarios where identifying high-income individuals is the priority.

End of Report